Forward Kinematics: Denavit-Hartenberg

For a serial robot arm, the transformation from frame i-1 to frame i:

$$T_i = \mathrm{Rot}(\theta, z) \cdot \mathrm{Trans}(d, z) \cdot \mathrm{Trans}(a, x) \cdot \mathrm{Rot}(\alpha, x)$$

Expanded as a 4×4 homogeneous transformation matrix:

$$\begin{bmatrix} \cos\theta & -\sin\theta\cos\alpha & \sin\theta\sin\alpha & a\cos\theta \\ \sin\theta & \cos\theta\cos\alpha & -\cos\theta\sin\alpha & a\sin\theta \\ 0 & \sin\alpha & \cos\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where:

- θ = joint angle (rotation about z-axis)
- d = link offset (translation along z-axis)
- a = link length (translation along x-axis)
- α = link twist (rotation about x-axis)

Overall arm transformation:

$$T_0^n = T_1 \cdot T_2 \cdot \ldots \cdot T_n$$

For an n-DOF serial manipulator, T_0^n gives the end-effector pose relative to the base.